



From Chinook Database to Business Intelligence Insights

Data Warehouse Design & Power BI Analysis Team Presentation

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Project Goals: Unlocking Data Potential

Database Transformation

Convert the Chinook operational database into a robust analytical system.

Enhanced Analysis

Enable in-depth sales and performance analysis for strategic decision-making.

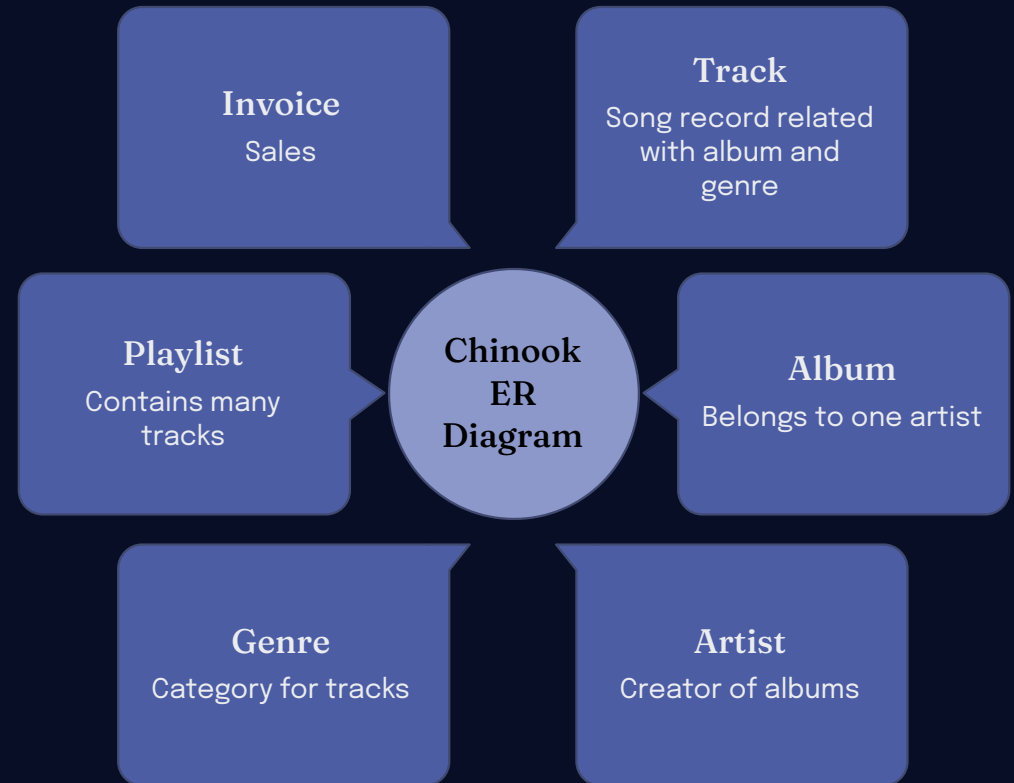
Power BI Integration

Leverage Power BI to generate actionable business insights and interactive dashboards.



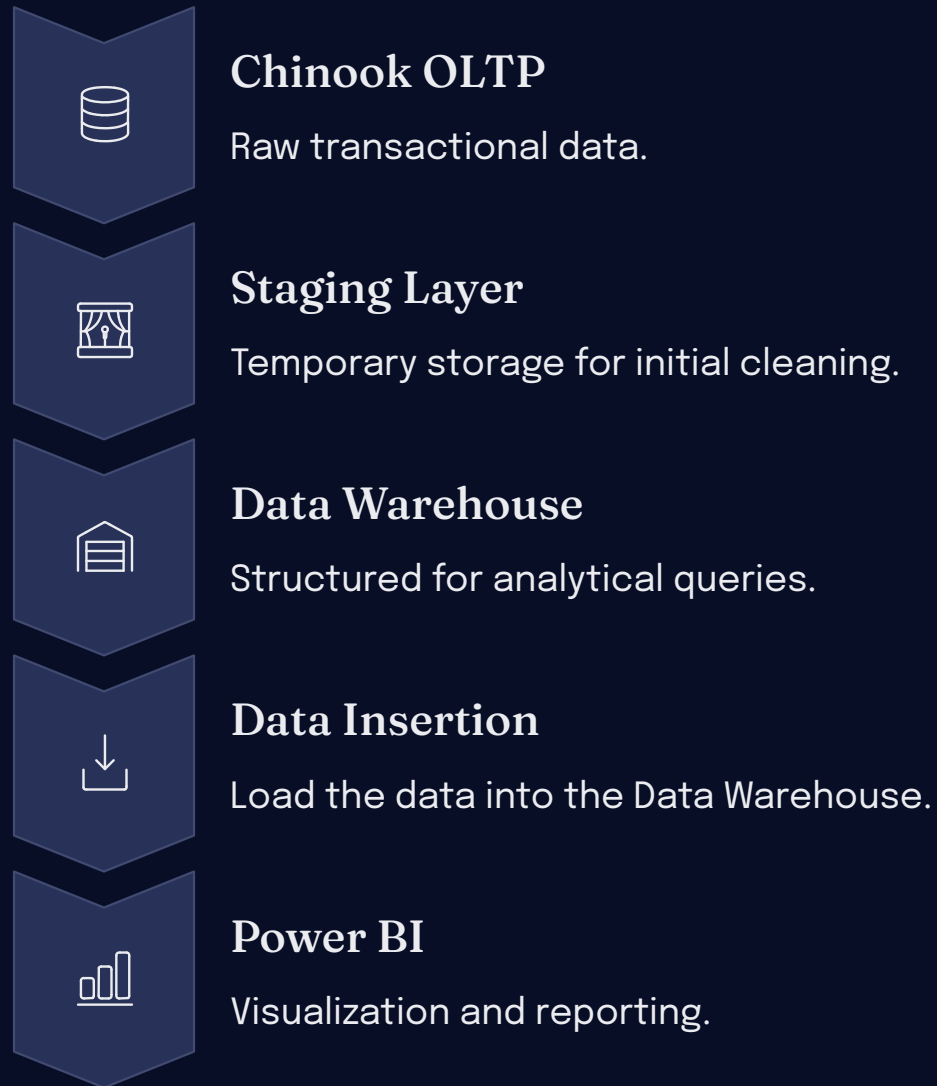
Chinook Database (OLTP)

- Each Track represents a song and belongs to one Album
- Each Album is associated with one Artist
- Tracks are categorized by Genre and may appear in multiple Playlists
- A Sale is represented by an Invoice
- Each Invoice may contain multiple tracks
- Each Invoice is issued to one Customer
- Each Customer is assigned a Support Employee



ETL & Data Preparation Overview

Our pipeline transforms raw transactional data into structured, analytical data.



Staging Layer: Intermediate Data Storage

The staging layer acts as a temporary, dedicated space for raw data, ensuring initial cleansing and transformation before it moves to the data warehouse.



Dedicated Staging Database

A separate database instance is used for all intermediate storage operations.



Employee Data Duplication

Only EmployeeId, FirstName, LastName, and Title were retained from the Chinook OLTP.



Customer Data Copy

The Customer table is also copied from Chinook DB without personal information of each customer.



Track Staging Table

Created by joining Track, Genre, Album, and Artist data.



Invoice Staging Table

Consolidates data from the Invoice and InvoiceLine tables.

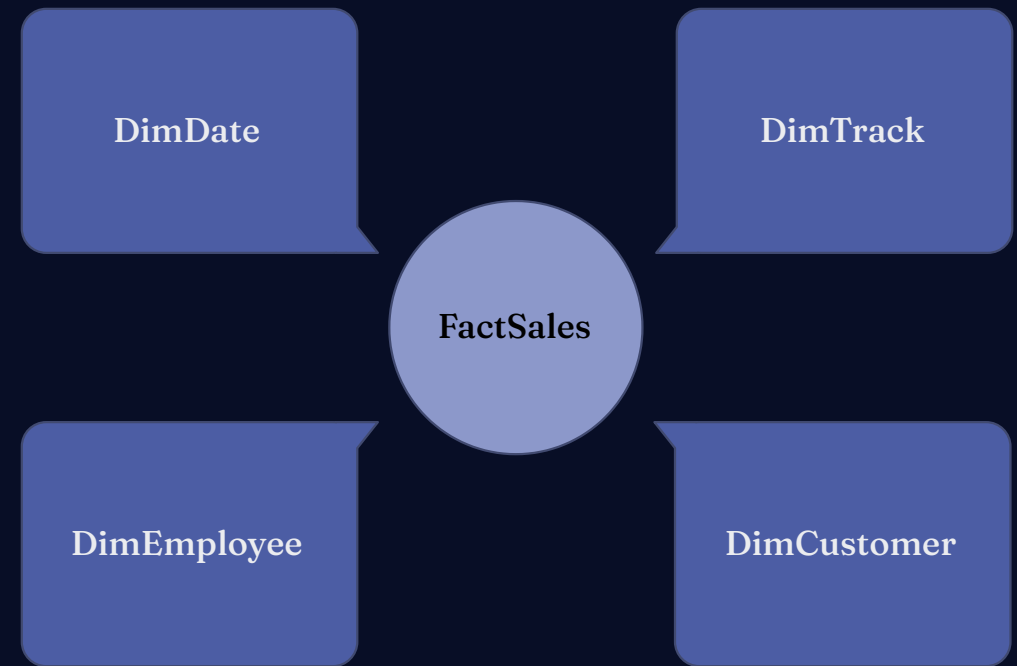


Attribute Preservation

All critical transactional attributes for sales analysis are preserved.

Data Warehouse (DW) Stage

- Created a FactSales table containing:
 - Foreign keys to Track, Customer, Employee, and SaleDate dimensions
 - InvoiceId, surrogate key of the Invoice initial table
 - FactSales measures such as unit price, and total amount
- Defined a DimTrack dimension table including:
 - TrackKey surrogate primary key
 - TrackID initial primary key
 - TrackName, Album, Artist, Composer and Genre attributes
 - Additional metadata columns for record validity and status
- Defined a DimCustomer dimension table with customer-related attributes
- Defined an DimEmployee dimension table for support representative data
- Created a DimDate dimension table with detailed calendar attributes



Data Quality & Insertion Logic



Handling Missing Values

Proactive strategies to identify and address incomplete data records.



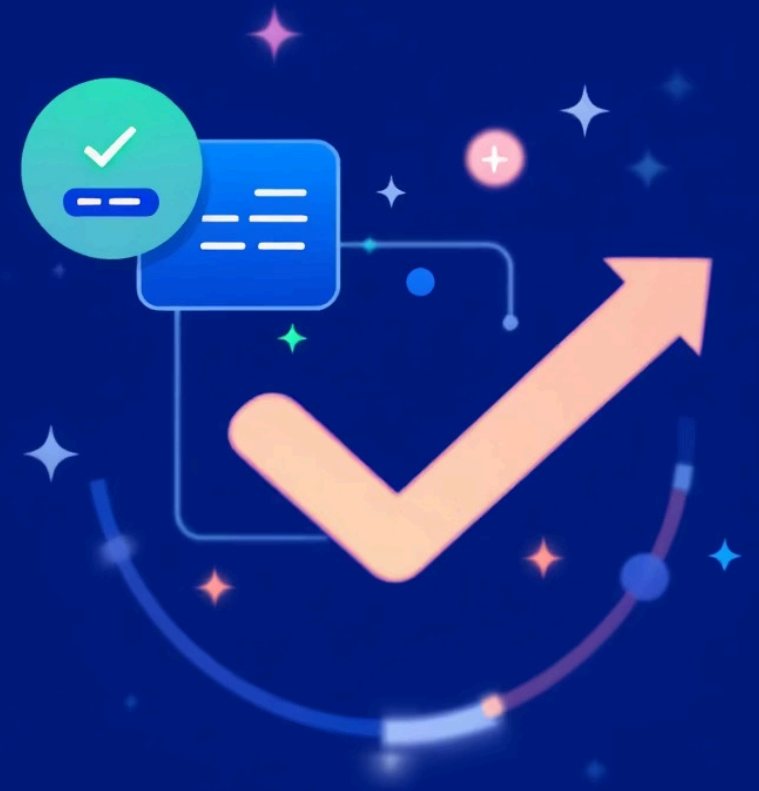
NULL Value Replacement

Systematic replacement of NULLs with 'N/A' or appropriate defaults for consistent reporting.



Date Dimension

Implementation of a dedicated Date dimension for robust time-based analysis and trend identification.

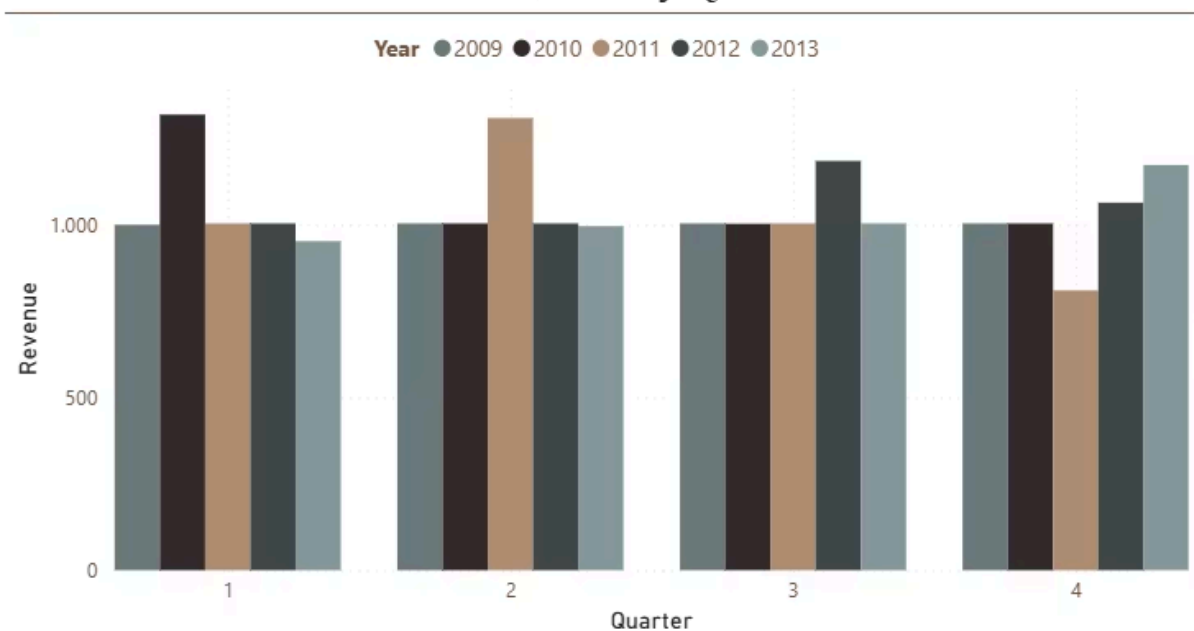




From Data to Insights

Power BI: Revenue Analysis

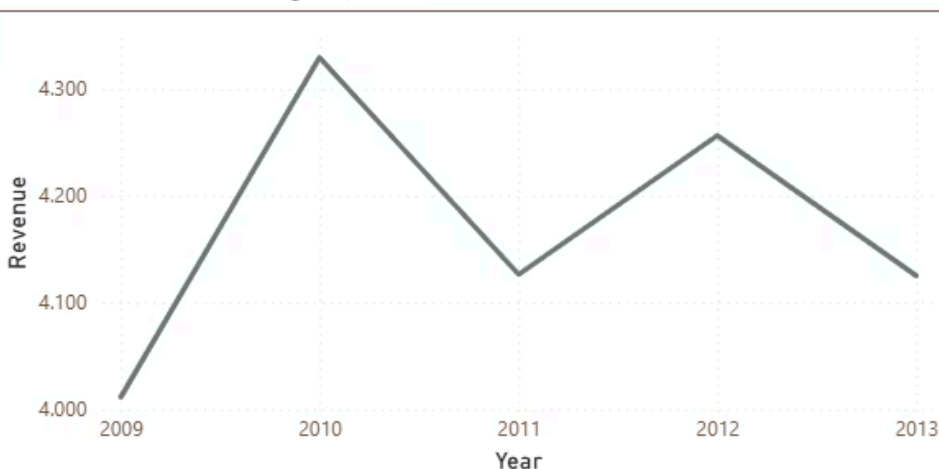
Total Revenue by Quarter



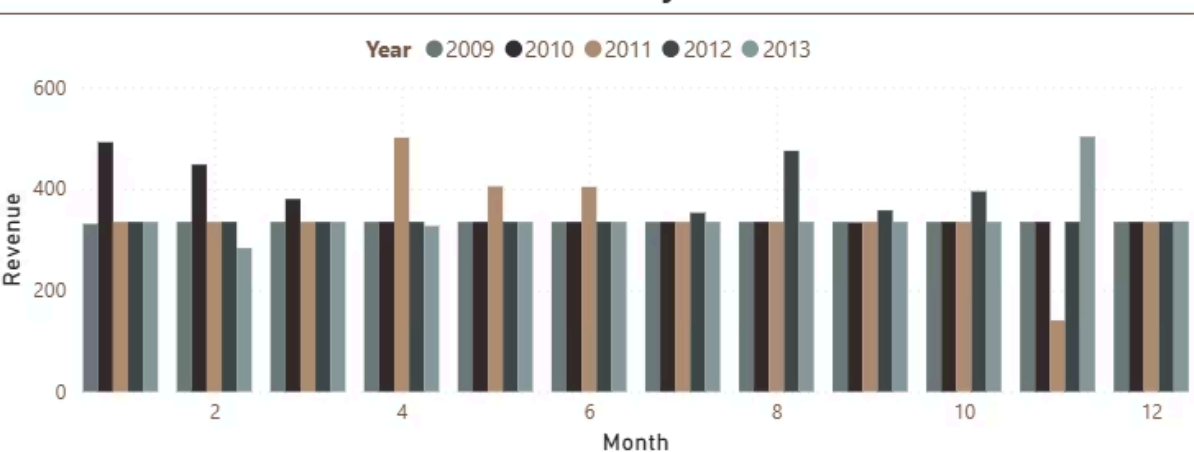
Year

- 2009
- 2010
- 2011
- 2012
- 2013

Total Revenue by Year



Total Revenue by Month



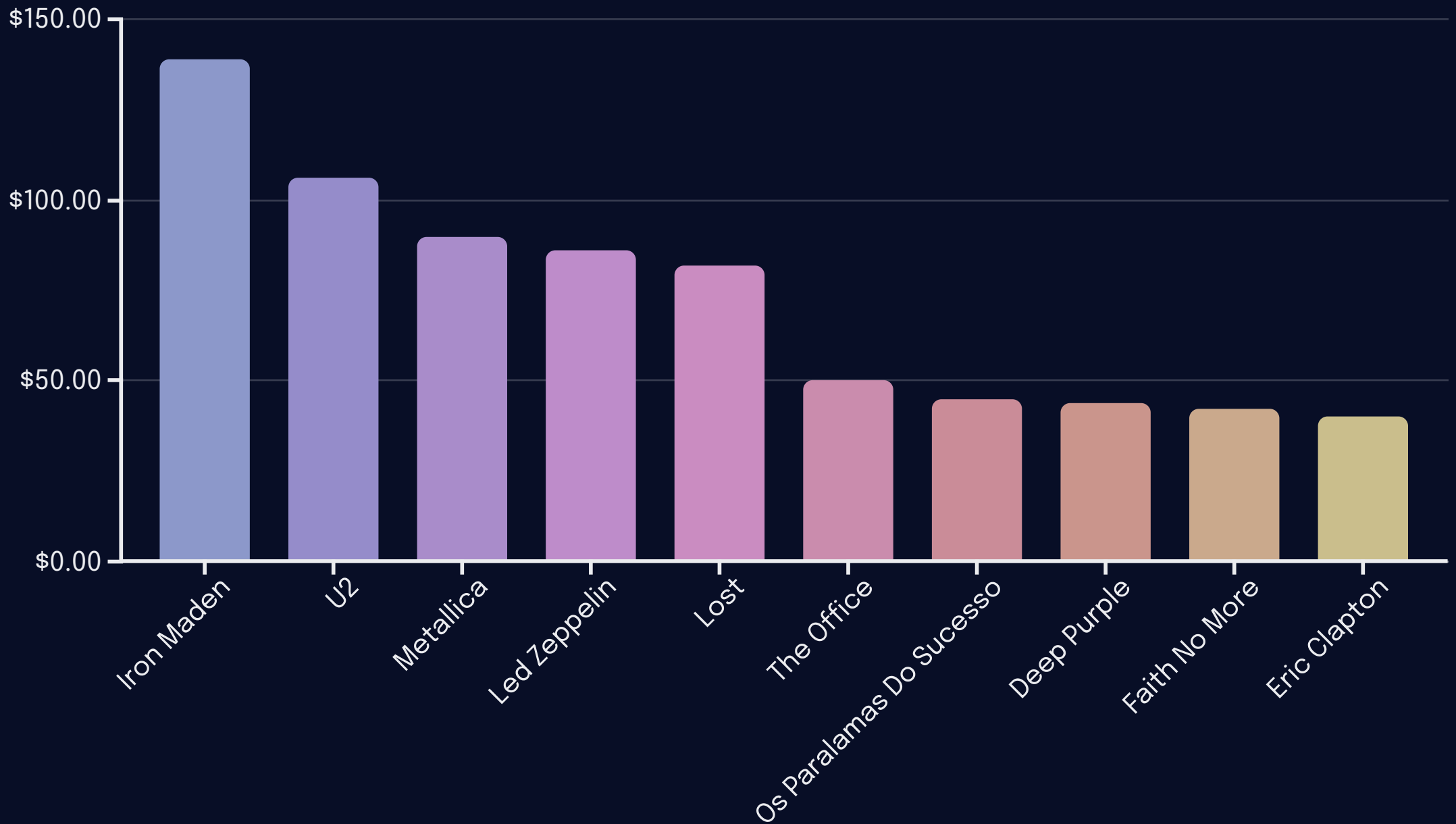
Sum of Total

Year

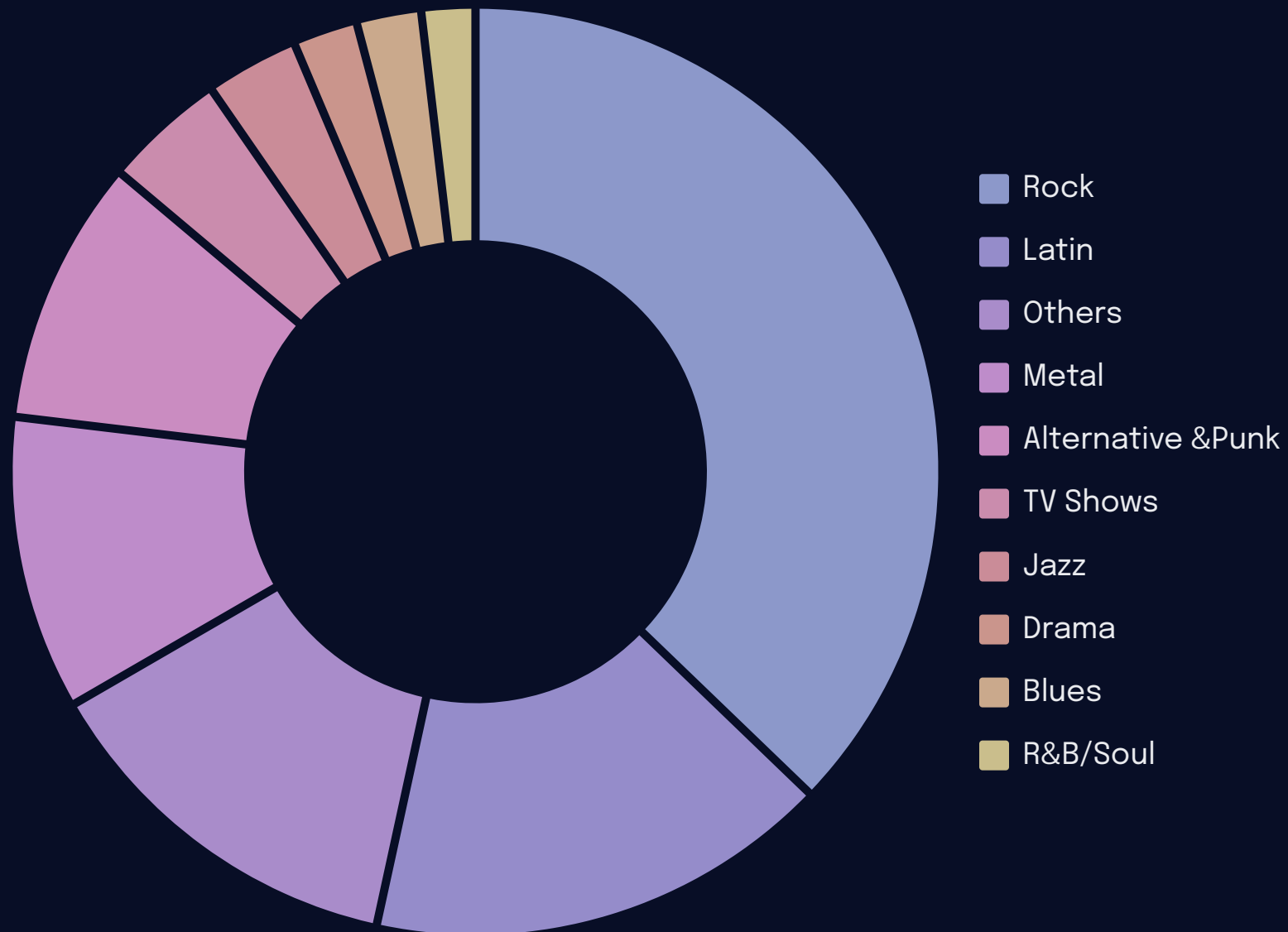
Percentage of Increase/Decrease

Sum of Total	Year	Percentage of Increase/Decrease
4.011,48	2009	0,00%
4.329,45	2010	7,93%
4.126,40	2011	-4,69%
4.256,25	2012	3,15%
4.125,04	2013	-3,08%

Power BI: Top 10 Artists by Sales

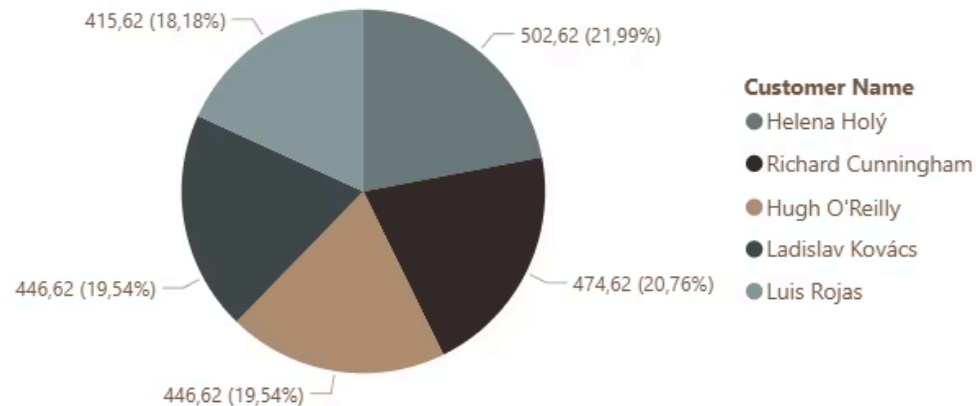


Power BI: Sales by Genre

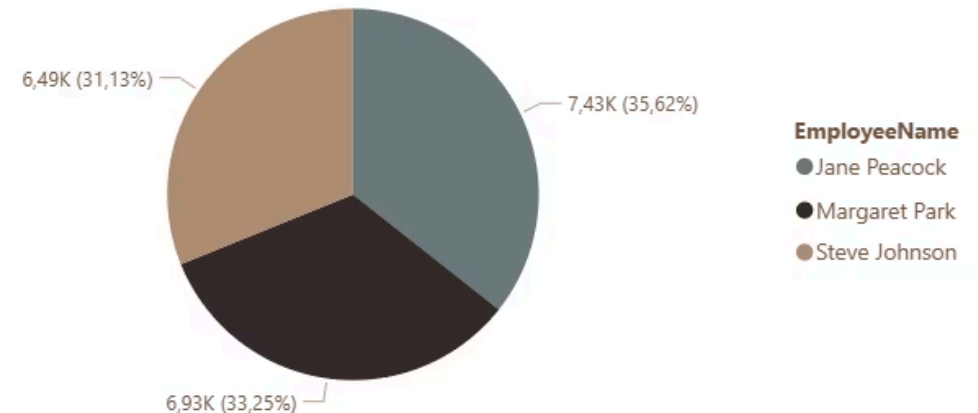


Power BI: Performance Analysis

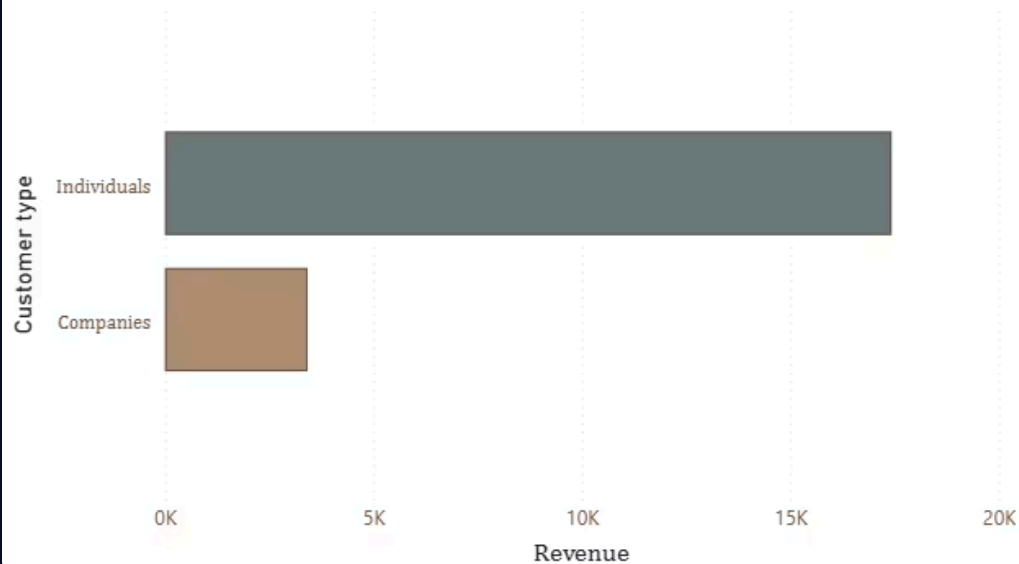
Top Customers



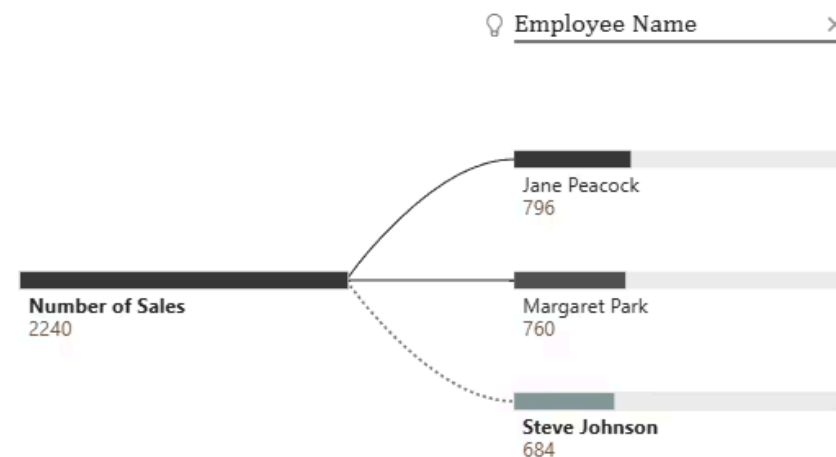
Top performing Employees



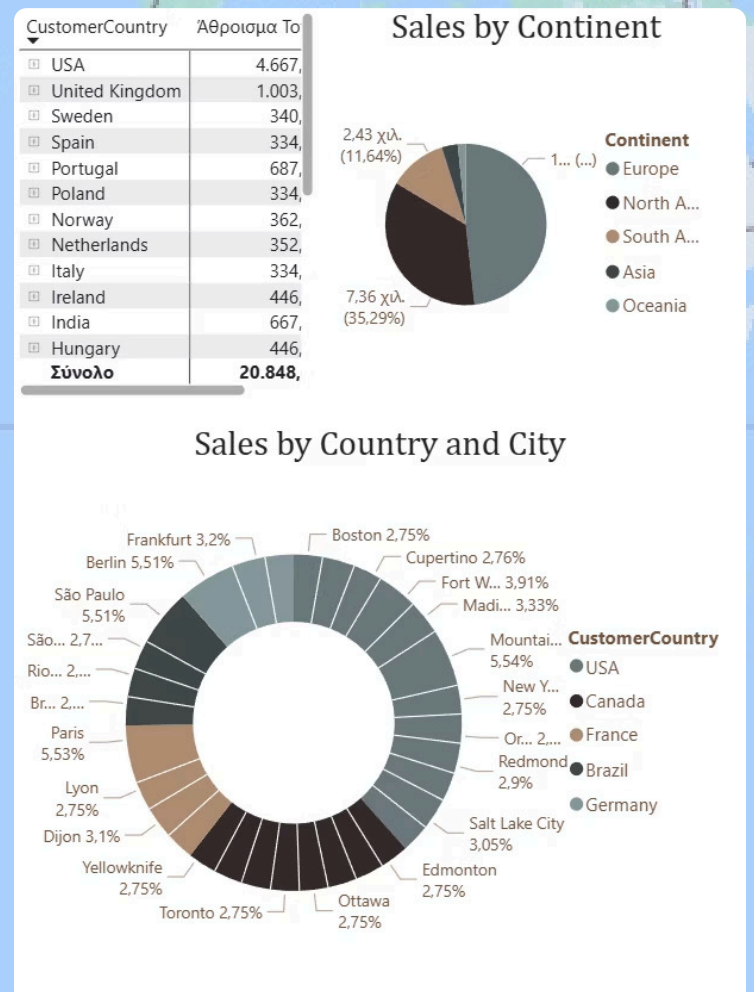
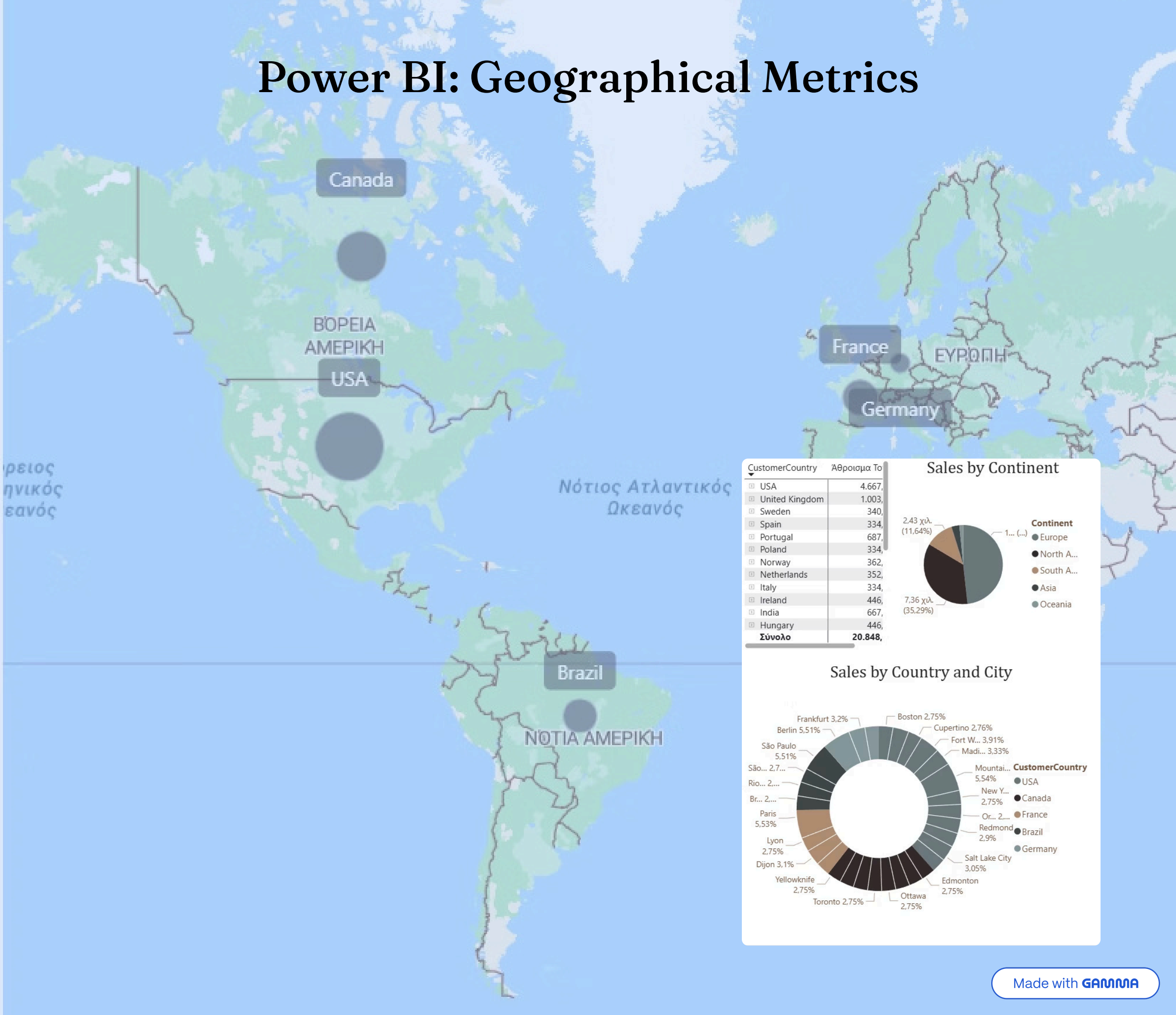
Sales by Customer Type



Number of Sales Per Employee



Power BI: Geographical Metrics



Project Summary:

Transactional Data to Business Intelligence

ETL Pipeline

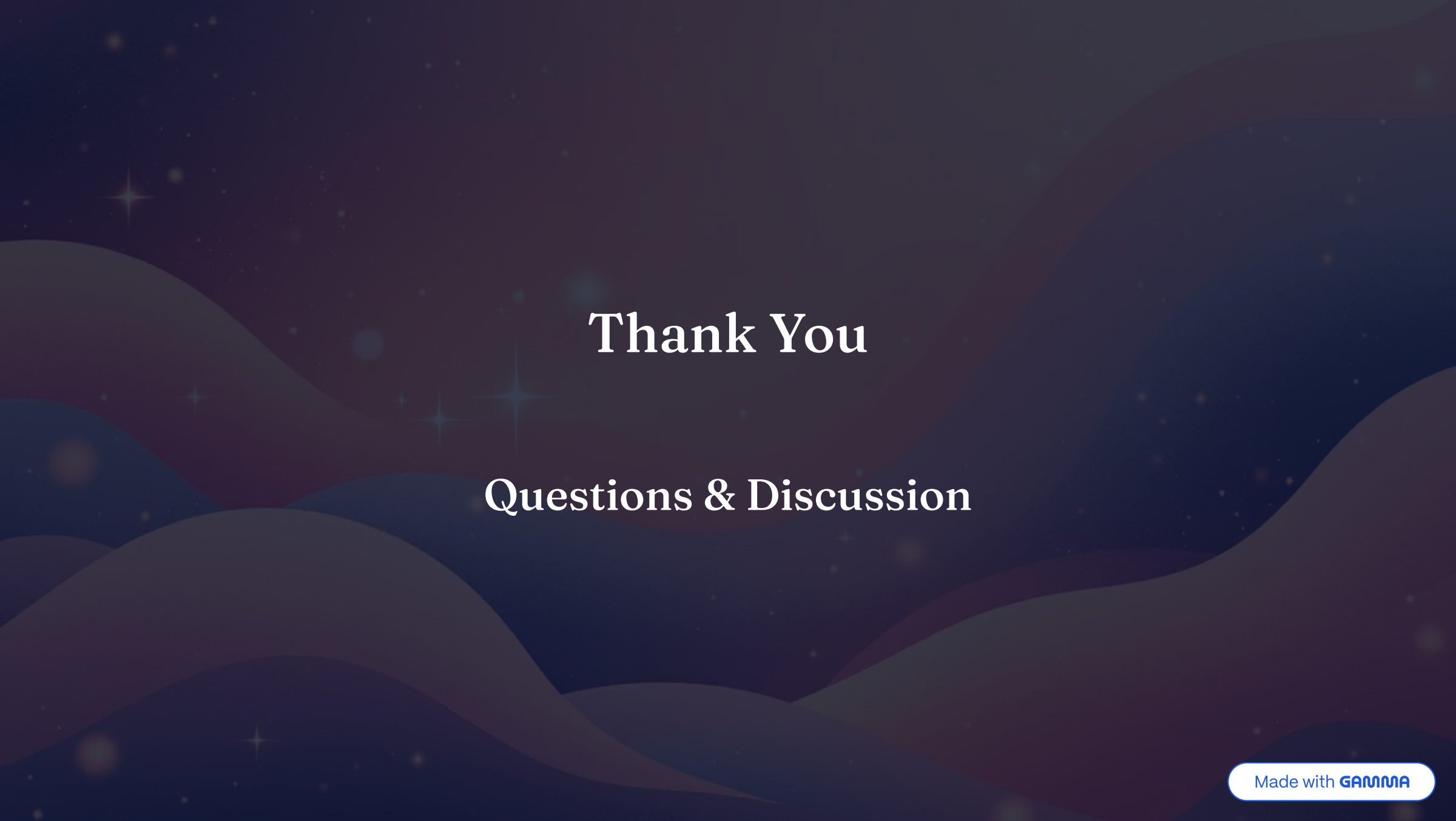
- Extracted raw data from Chinook OLTP database
- Staged data for cleaning and transformation
- Loaded into structured Data Warehouse

Data Warehouse Architecture

- Built star schema
- Implemented surrogate keys
- Optimized for analytical queries

Business Intelligence Outcomes

- Revenue analysis by geography and genre
- Performance metrics and sales trends
- Top-performing albums and artists
- Actionable insights for strategic decisions



Thank You

Questions & Discussion